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Experiment 1

Aim: Version Control Operations using Git

Lab Outcome: Mapped LO2

LO2: Use Git effectively for version control, source code management and collaborative team development.

Theory:

Git is a distributed version control system that enables developers to track changes in source code, manage multiple versions of a project, and collaborate efficiently with other developers. It allows every developer to maintain a complete copy of the repository, making development faster, reliable, and independent of network connectivity. Version control is an essential practice in modern software engineering as it records every modification made to a project. Git stores project snapshots as commits, each identified by a unique commit hash, thereby maintaining a complete history of the project.

In this project, a local Git repository was created and connected to a remote GitHub repository. Various Git commands were executed to understand the complete version control workflow. Repository status was monitored using git status, project files were staged using git add ., and changes were permanently recorded using git commit. The project history was examined through git log, while differences between file versions were analyzed using git diff. The local repository was synchronized with GitHub using git push, and remote repository details were verified using git remote and git ls-remote origin.

Branch management was performed by creating, switching, and deleting branches to demonstrate isolated feature development without affecting the

main branch. Repository cloning was also carried out to understand how an existing remote repository can be copied to another system while preserving its complete commit history and configuration.

The experiment further explored collaboration by adding a collaborator to the GitHub repository, illustrating how multiple developers can contribute to the same project. Rollback operations were demonstrated using git restore, which discards uncommitted changes and restores files to their previously committed state. Additionally, the git revert command was performed to safely undo a committed change by creating a new commit that reverses the selected commit while preserving the repository history.

Overall, Git provides an efficient mechanism for source code management by supporting version tracking, branching, collaboration, rollback, and synchronization with remote repositories. These features make Git an indispensable tool for Agile Software Development and DevOps workflows, ensuring code integrity, traceability, and effective team collaboration throughout the software development lifecycle.

Commands:

Sr. No.	Git Command	Description
1	dir	Displays the files and folders present in the current working directory.
2	git init	Initializes a new local Git repository in the current directory.
3	git status	Displays the current state of the repository, including modified, staged, and untracked files.
4	git add .	Stages all new and modified files for the next commit.
5	git commit -m "Initial calculator project"	Creates the first snapshot of the project and saves it in the repository history.
6	git commit -m "Added square function"	Records the addition of the square function as a separate version of the project.
7	git commit -m "Added rollback demo"	Commits the changes made to demonstrate rollback and revert operations.
8	git log	Displays the complete commit history with commit hashes, author, date, and messages.
9	git log --oneline	Displays a compact one-line summary of the commit history.
10	git diff	Displays the differences between the working directory and the last committed version of files.
11	git remote add origin <repository-url>	Connects the local repository to a remote GitHub repository.
12	git remote -v	Displays the fetch and push URLs of the configured remote repository.
13	git ls-remote origin	Lists all references (branches and commit hashes) available in the remote repository.
14	git push -u origin main	Uploads the local repository to GitHub and establishes tracking between local and remote branches.

15	git push	Pushes subsequent commits from the local repository to GitHub.
16	git merge feature-square	Merges the changes from the feature-square branch into the main branch
17	git branch	Displays all local branches or creates a new branch when a name is specified.
18	git branch feature-square	Creates a new branch named feature-square without switching to it.
19	git switch feature-square	Switches from the current branch to the feature branch.
20	git switch main	Returns from the feature branch to the main branch.
21	git branch -d feature-square	Deletes the specified branch after it has been merged or is no longer required.
22	git config --list	Displays the Git configuration settings such as username, email, remotes, and repository information.
23	git restore calculator.py	Discards uncommitted changes made to the specified file, restoring its last committed version (rollback).
24	git show	Displays detailed information about the latest commit by default.
25	git clone <repository-url>	Creates a complete local copy of an existing remote Git repository.
26	git pull origin main	Downloads and merges the latest changes from the remote repository into the local repository.
27	git fetch	Downloads updates from the remote repository without merging them into the local branch.
28	git remote show origin	Displays detailed information about the remote repository, including tracked branches and URLs.
29	git config --list	Displays all Git configuration settings currently in use

Output:



```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\Kumud\Desktop\Library Management>
PS C:\Users\Kumud\Desktop\Library Management> cd "C:\Users\Kumud\Desktop\Library Management"
>>
>> Remove-Item -Recurse -Force "python-calculator" -ErrorAction SilentlyContinue
>> Remove-Item -Recurse -Force "python-calculator-clone" -ErrorAction SilentlyContinue
>>
>> dir


    Directory: C:\Users\Kumud\Desktop\Library Management


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        11-04-2026  11:31 PM                backend
d-----        11-04-2026  10:36 PM                frontend
-a----        11-04-2026  11:39 PM             58 .gitigno
                                                  re
-a----        17-04-2026  03:53 PM           3510 README.m
                                                  d

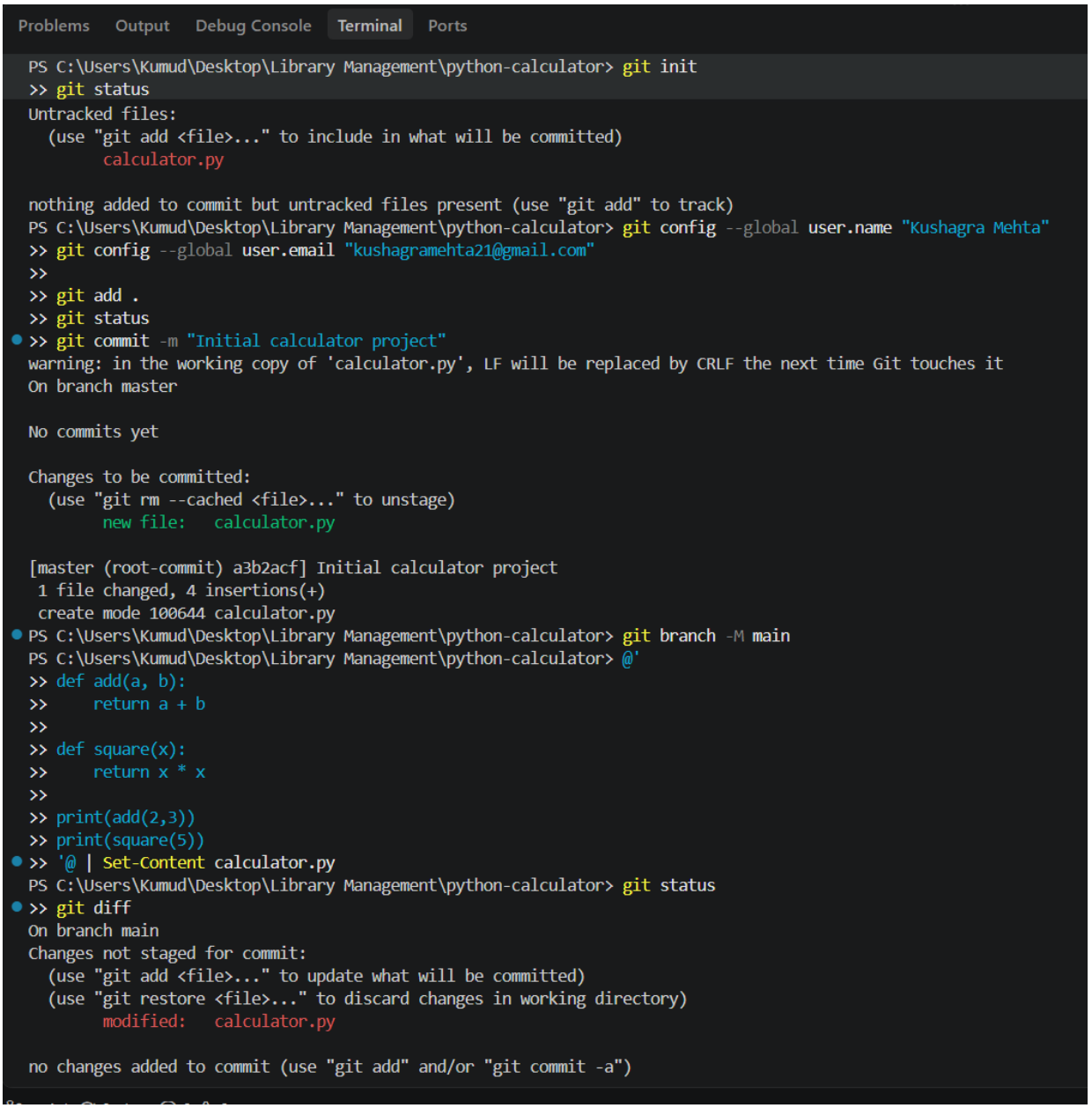
                                          mkdir python-calculator
>> cd python-calculatorop\Library Management>


    Directory: C:\Users\Kumud\Desktop\Library Management


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        31-07-2026  02:11 PM                python-calculator


PS C:\Users\Kumud\Desktop\Library Management\python-calculator> @'
>> def add(a, b):
>>     return a + b
>>
>> print(add(2,3))
>> '@ | Set-Content calculator.py
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> git init
>> git status
Initialized empty Git repository in C:/Users/Kumud/Desktop/Library Management/python-calculator/.git/
On branch master

No commits yet
```


Problems
Output
Debug Console
Terminal
Ports
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> git init
>> git status
Untracked files:
(use "git add <file>..." to include in what will be committed)
calculator.py
nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> git config --global user.name "Kushagra Mehta"
>> git config --global user.email "kushagramehta21@gmail.com"
>>
>> git add .
>> git status
>> git commit -m "Initial calculator project"
warning: in the working copy of 'calculator.py', LF will be replaced by CRLF the next time Git touches it
On branch master
No commits yet
Changes to be committed:
(use "git rm --cached <file>..." to unstage)
new file: calculator.py
[master (root-commit) a3b2acf] Initial calculator project
1 file changed, 4 insertions(+)
create mode 100644 calculator.py
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> git branch -M main
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> @'
>> def add(a, b):
>> return a + b
>>
>> def square(x):
>> return x * x
>>
>> print(add(2,3))
>> print(square(5))
>> '@ | Set-Content calculator.py
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> git status
>> git diff
On branch main
Changes not staged for commit:
(use "git add <file>..." to update what will be committed)
(use "git restore <file>..." to discard changes in working directory)
modified: calculator.py
no changes added to commit (use "git add" and/or "git commit -a")

```
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> git status
>> git diff
warning: in the working copy of 'calculator.py', LF will be replaced by CRLF the next time Git touches it
diff --git a/calculator.py b/calculator.py
index 50d6420..a263267 100644
--- a/calculator.py
+++ b/calculator.py
@@ -1,4 +1,8 @@
 def add(a, b):
     return a + b

+def square(x):
+    return x * x
+
 print(add(2,3))
+print(square(5))
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> git add .
>> git commit -m "Added square function"
warning: in the working copy of 'calculator.py', LF will be replaced by CRLF the next time Git touches it
[main 938f346] Added square function
 1 file changed, 4 insertions(+)
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> git log
>> git log --oneline
>> git show
commit 938f346d6fc2916ad0c603a5861d58cbf8730ace (HEAD -> main)
Author: Kushagra Mehta <kushagramehta21@gmail.com>
Date:   Fri Jul 31 14:12:03 2026 +0530

    Added square function

commit a3b2acf3c95e669c3899af205cc62a2195a10db2
Author: Kushagra Mehta <kushagramehta21@gmail.com>
Date:   Fri Jul 31 14:11:50 2026 +0530

    Initial calculator project
938f346 (HEAD -> main) Added square function
a3b2acf Initial calculator project
commit 938f346d6fc2916ad0c603a5861d58cbf8730ace (HEAD -> main)
Author: Kushagra Mehta <kushagramehta21@gmail.com>
Date:   Fri Jul 31 14:12:03 2026 +0530

    Added square function

diff --git a/calculator.py b/calculator.py
index 50d6420..a263267 100644
```

```
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> git log
>> git log --oneline
>> git show
+++ b/calculator.py
@@ -1,4 +1,8 @@
 def add(a, b):
     return a + b

+def square(x):
+    return x * x
+
 print(add(2,3))
+print(square(5))
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> git remote add origin https://github.com/kushagra819/python-calculator.git
>> git remote -v
>> git ls-remote origin
origin  https://github.com/kushagra819/python-calculator.git (fetch)
origin  https://github.com/kushagra819/python-calculator.git (push)
69dc1d95c23f67e6a10f5b53a675316885c0846d        HEAD
69dc1d95c23f67e6a10f5b53a675316885c0846d        refs/heads/main
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> git push -u origin main --force
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 20 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 577 bytes | 577.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/kushagra819/python-calculator.git
 + 69dc1d9...938f346 main -> main (forced update)
branch 'main' set up to track 'origin/main'.
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> git branch feature-square
>> git branch
>> git switch feature-square
>> git branch
  feature-square
* main
Switched to branch 'feature-square'
* feature-square
  main
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> @'
>> def add(a, b):
>>     return a + b
>>
>> def square(x):
>>     return x * x
```

```
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> git add .
>> git commit -m "Added cube function"
warning: in the working copy of 'calculator.py', LF will be replaced by CRLF the next time Git touches it
[feature-square 6a23c66] Added cube function
 1 file changed, 4 insertions(+)
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> git switch main
>> git merge feature-square
>> git branch -d feature-square
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
Updating 938f346..6a23c66
Fast-forward
 calculator.py | 4 ++++
 1 file changed, 4 insertions(+)
Deleted branch feature-square (was 6a23c66).
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> @'
>> def add(a, b):
>>     return a + b
>>
>> def square(x):
>>     return x * x
>>
>> def cube(x):
>>     return x * x * x
>>
>> print(add(2,3))
>> print(square(5))
>> print(cube(3))
>>
>> print("Wrong Code")
>> '@ | Set-Content calculator.py
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> git diff
>> git restore calculator.py
>> git status
warning: in the working copy of 'calculator.py', LF will be replaced by CRLF the next time Git touches it
diff --git a/calculator.py b/calculator.py
index 78ef276..3bb4605 100644
--- a/calculator.py
+++ b/calculator.py
@@ -10,3 +10,5 @@ def cube(x):
 print(add(2,3))
 print(square(5))
 print(cube(3))
+
+print("Wrong Code")

main* 0↓ 1↑  ⊗ 0 ⚠ 0
```

```
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> git diff
>> git restore calculator.py
>> git status
nothing to commit, working tree clean
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> @'
>> def add(a, b):
>>     return a + b
>>
>> def square(x):
>>     return x * x
>>
>> def cube(x):
>>     return x * x * x
>>
>> print(add(2,3))
>> print(square(5))
>> print(cube(3))
>>
>> print("Rollback Demo")
>> '@ | Set-Content calculator.py
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> git add .
>> git commit -m "Added rollback demo"
warning: in the working copy of 'calculator.py', LF will be replaced by CRLF the next time Git touches it
[main a653e52] Added rollback demo
 1 file changed, 2 insertions(+)
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> git add .
>> git commit -m "Added rollback demo"
On branch main
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> git log --oneline
a653e52 (HEAD -> main) Added rollback demo
6a23c66 Added cube function
938f346 (origin/main) Added square function
a3b2acf Initial calculator project
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> git revert abc1234
fatal: bad revision 'abc1234'
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> git log --oneline
a653e52 (HEAD -> main) Added rollback demo
6a23c66 Added cube function
938f346 (origin/main) Added square function
a3b2acf Initial calculator project
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> git fetch
```

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\Kumud\Desktop\Library Management> dir

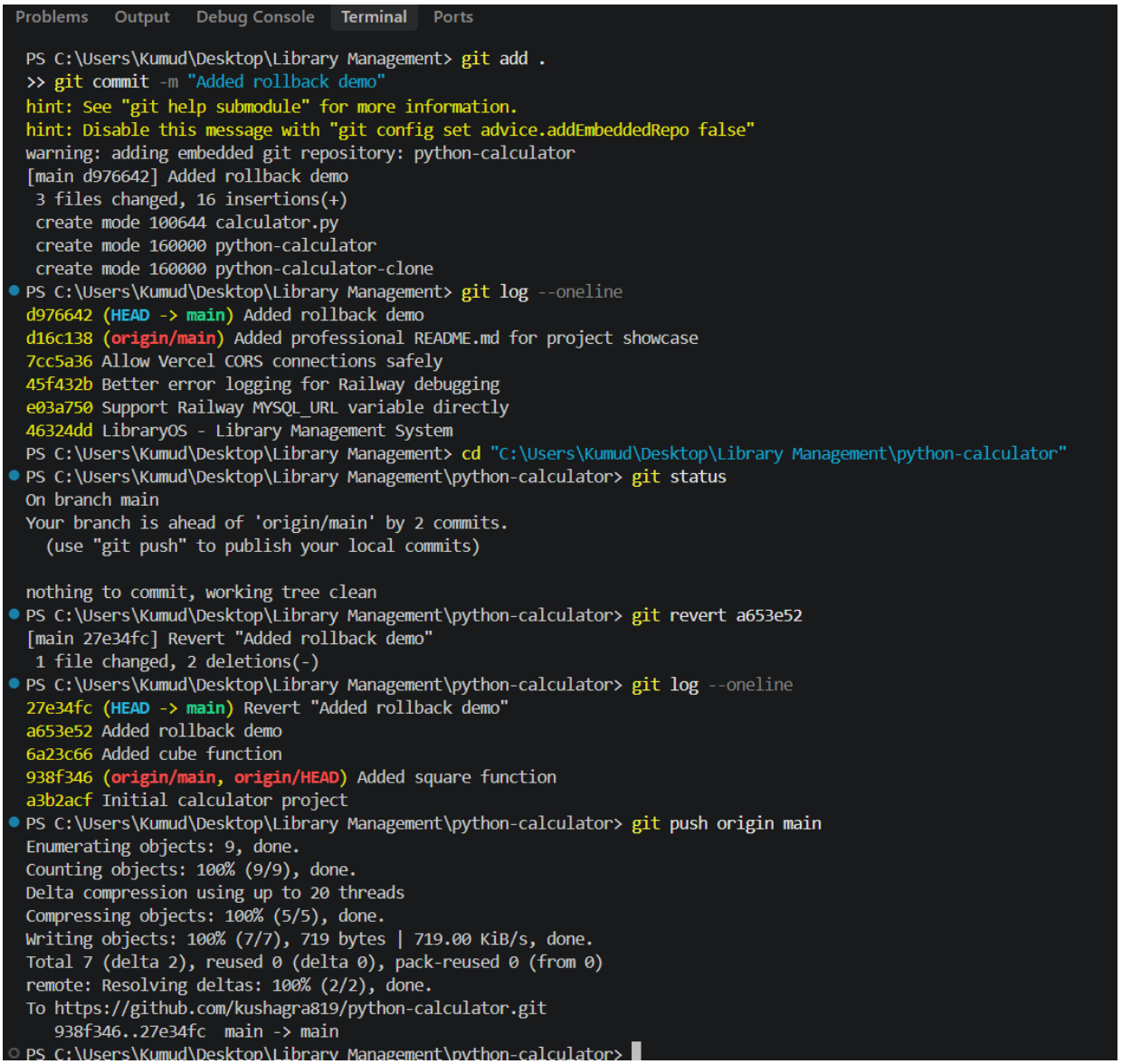

    Directory: C:\Users\Kumud\Desktop\Library Management


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        11-04-2026  11:31 PM                backend
d-----        11-04-2026  10:36 PM                frontend
d-----        31-07-2026  02:13 PM                python-calculator
d-----        31-07-2026  02:14 PM                python-calculator-clone
-a----        11-04-2026  11:39 PM             58 .gitignore
-a----        17-04-2026  03:53 PM           3510 README.md


PS C:\Users\Kumud\Desktop\Library Management> @'
>> def add(a, b):
>>     return a + b
>>
>> def square(x):
>>     return x * x
>>
>> def cube(x):
>>     return x * x * x
>>
>> print(add(2,3))
>> print(square(5))
>> print(cube(3))
>>
>> print("Wrong Code")
>> '@ | Set-Content calculator.py
PS C:\Users\Kumud\Desktop\Library Management> git diff
PS C:\Users\Kumud\Desktop\Library Management> git restore calculator.py
error: pathspec 'calculator.py' did not match any file(s) known to git
PS C:\Users\Kumud\Desktop\Library Management> git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        calculator.py
        python-calculator-clone/
        python-calculator/
```


Problems Output Debug Console Terminal Ports
PS C:\Users\Kumud\Desktop\Library Management> git status
PS C:\Users\Kumud\Desktop\Library Management> @'
>> def add(a, b):
>>   return a + b
>>
>> def square(x):
>>   return x * x
>>
>> def cube(x):
>>   return x * x * x
>>
>> print(add(2,3))
>> print(square(5))
>> print(cube(3))
>>
>> print("Rollback Demo")
>> '@ | Set-Content calculator.py
PS C:\Users\Kumud\Desktop\Library Management> git add .
>> git commit -m "Added rollback demo"
warning: in the working copy of 'calculator.py', LF will be replaced by CRLF the next time Git touches it
warning: adding embedded git repository: python-calculator-clone
hint: You've added another git repository inside your current repository.
hint: Clones of the outer repository will not contain the contents of
hint: the embedded repository and will not know how to obtain it.
hint: If you meant to add a submodule, use:
hint:
hint: git submodule add <url> python-calculator-clone
hint:
hint: If you added this path by mistake, you can remove it from the
hint: index with:
hint:
hint: git rm --cached python-calculator-clone
hint:
hint: See "git help submodule" for more information.
hint: Disable this message with "git config set advice.addEmbeddedRepo false"
warning: adding embedded git repository: python-calculator
[main d976642] Added rollback demo
 3 files changed, 16 insertions(+)
 create mode 100644 calculator.py
 create mode 160000 python-calculator
 create mode 160000 python-calculator-clone
PS C:\Users\Kumud\Desktop\Library Management> git log --oneline
d976642 (HEAD -> main) Added rollback demo
d16c138 (origin/main) Added professional README.md for project showcase
7cc5a36 Allow Vercel CORS connections safely


Problems Output Debug Console Terminal Ports
PS C:\Users\Kumud\Desktop\Library Management> git add .
>> git commit -m "Added rollback demo"
hint: See "git help submodule" for more information.
hint: Disable this message with "git config set advice.addEmbeddedRepo false"
warning: adding embedded git repository: python-calculator
[main d976642] Added rollback demo
3 files changed, 16 insertions(+)
create mode 100644 calculator.py
create mode 160000 python-calculator
create mode 160000 python-calculator-clone
PS C:\Users\Kumud\Desktop\Library Management> git log --oneline
d976642 (HEAD -> main) Added rollback demo
d16c138 (origin/main) Added professional README.md for project showcase
7cc5a36 Allow Vercel CORS connections safely
45f432b Better error logging for Railway debugging
e03a750 Support Railway MYSQL_URL variable directly
46324dd LibraryOS - Library Management System
PS C:\Users\Kumud\Desktop\Library Management> cd "C:\Users\Kumud\Desktop\Library Management\python-calculator"
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> git status
On branch main
Your branch is ahead of 'origin/main' by 2 commits.
(use "git push" to publish your local commits)
nothing to commit, working tree clean
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> git revert a653e52
[main 27e34fc] Revert "Added rollback demo"
1 file changed, 2 deletions(-)
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> git log --oneline
27e34fc (HEAD -> main) Revert "Added rollback demo"
a653e52 Added rollback demo
6a23c66 Added cube function
938f346 (origin/main, origin/HEAD) Added square function
a3b2acf Initial calculator project
PS C:\Users\Kumud\Desktop\Library Management\python-calculator> git push origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 20 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (7/7), 719 bytes | 719.00 KiB/s, done.
Total 7 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To https://github.com/kushagra819/python-calculator.git
938f346..27e34fc main -> main
PS C:\Users\Kumud\Desktop\Library Management\python-calculator>

Conclusion:

The experiment successfully demonstrated the fundamental operations of Git as a distributed version control system used in Agile Software Development and DevOps. Various Git commands such as git status, git add, git commit, git log, git diff, git push, git clone, git remote, git ls-remote, and git config were executed to understand how Git tracks, stores, and synchronizes project changes.

This experiment was instrumental in helping gain an understanding of Git's role in maintaining project integrity, facilitating collaborative development, and supporting the version control practices essential in Agile Software Development and DevOps environments. The experiment improved my real-world understanding of Git and its significance in modern software engineering workflows.